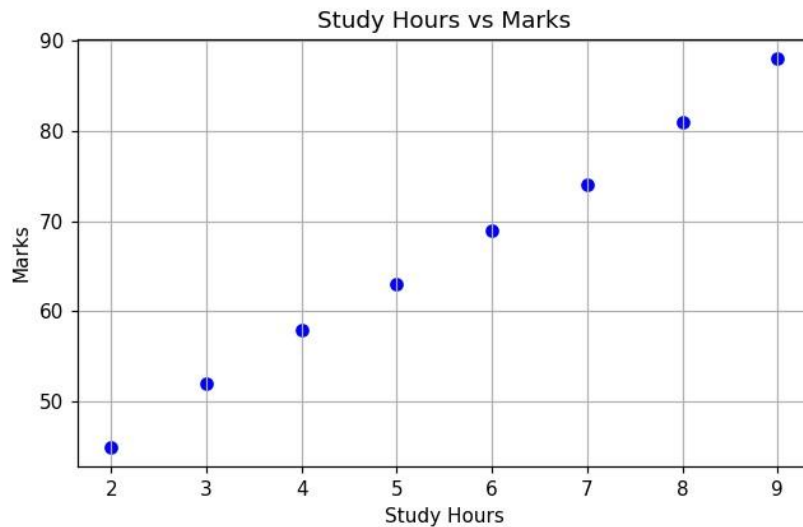


Q1. Study Hours vs Marks (R)

```
StudyHours <- c(2,3,4,5,6,7,8,9) Marks  
<- c(45,52,58,63,69,74,81,88) df <-  
data.frame(StudyHours, Marks)
```

```
plot(df$StudyHours, df$Marks,      main =  
"Study Hours vs Marks",      xlab = "Study  
Hours", ylab = "Marks",      pch = 19, col  
= "blue")
```

```
cor(df$StudyHours, df$Marks)
```

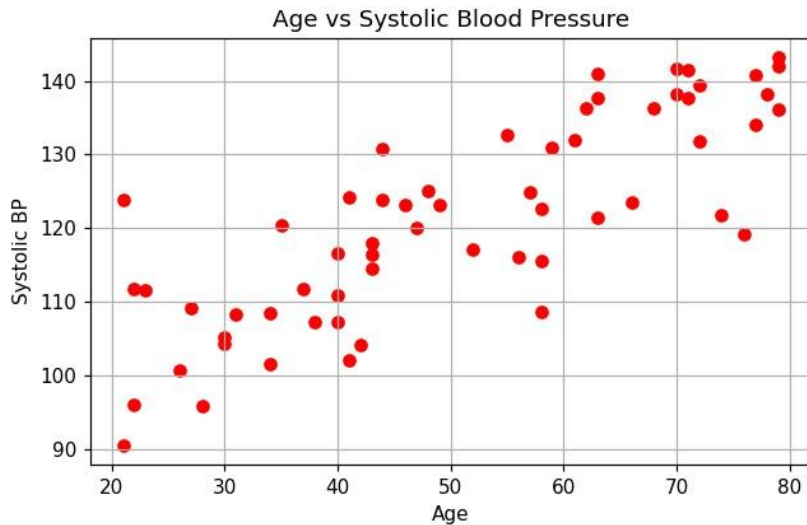


Q2. Age vs Systolic Blood Pressure (Python)

```
import matplotlib.pyplot as plt import  
numpy as np
```

```
age = np.random.randint(20, 80, 60)  
bp = 90 + 0.6*age + np.random.normal(0, 8, 60)
```

```
plt.scatter(age, bp, color='red') plt.grid(True)  
plt.title("Age vs Systolic Blood Pressure") plt.xlabel("Age") plt.ylabel("Systolic BP")  
plt.show()
```

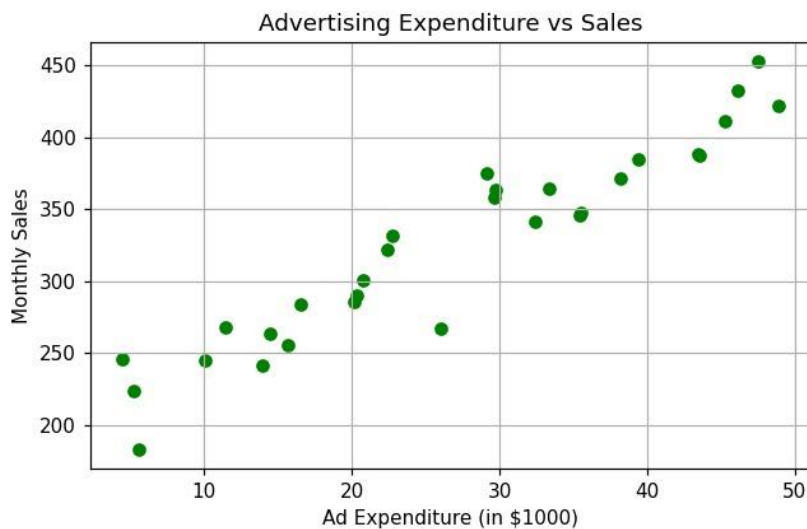


Q3. Advertising Expenditure vs Sales

```
import matplotlib.pyplot as plt
import numpy as np
```

```
ad = np.linspace(5, 50, 30) + np.random.normal(0, 2, 30)
sales = 200 + 4.5*ad + np.random.normal(0, 20, 30)
```

```
plt.scatter(ad, sales, color='green')
plt.title("Advertising Expenditure vs Sales")
plt.xlabel("Ad Expenditure ($000)")
plt.ylabel("Monthly Sales")
plt.grid(True)
plt.show()
```

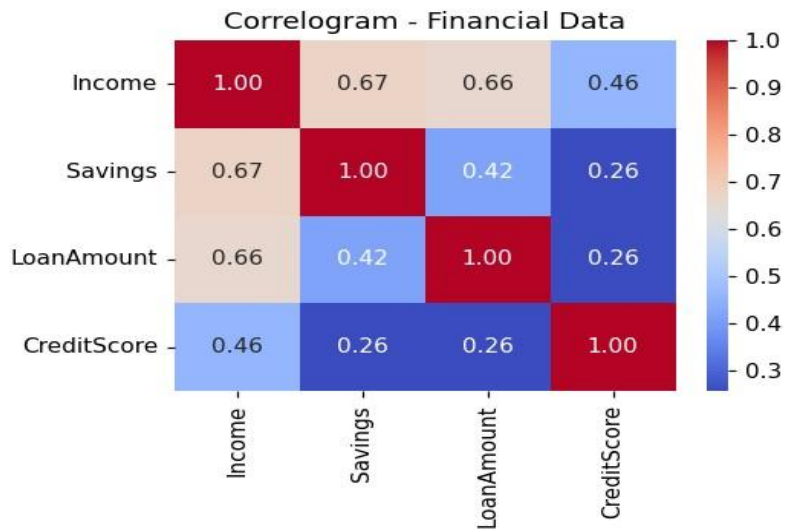


Q4 & Q5. Financial Data — Correlation Matrix & Correlogram

```
import pandas as pd, seaborn as sns, matplotlib.pyplot as plt
```

```
df = pd.DataFrame({'Income':income,'Savings':savings,
'LoanAmount':loan,'CreditScore':credit})
corr = df.corr()
```

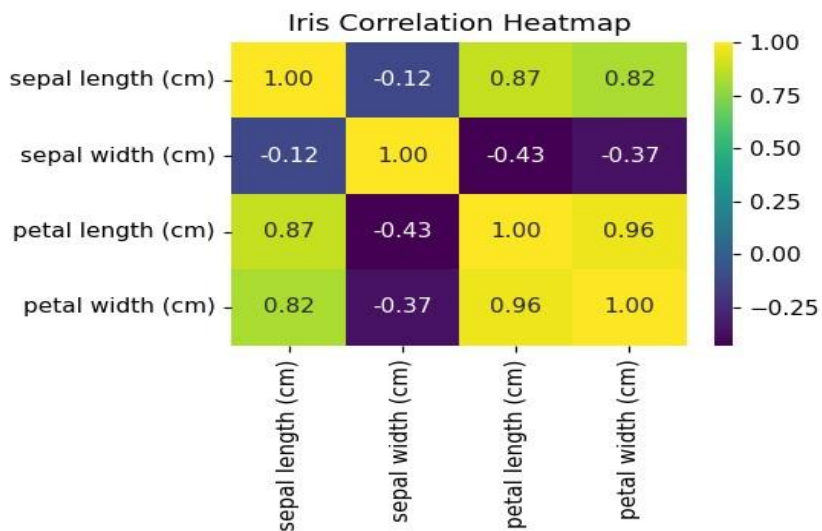
```
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f") plt.title("Correlogram - Financial Data") plt.show()
```



Q6. Iris Dataset — Correlation Heatmap (Seaborn)

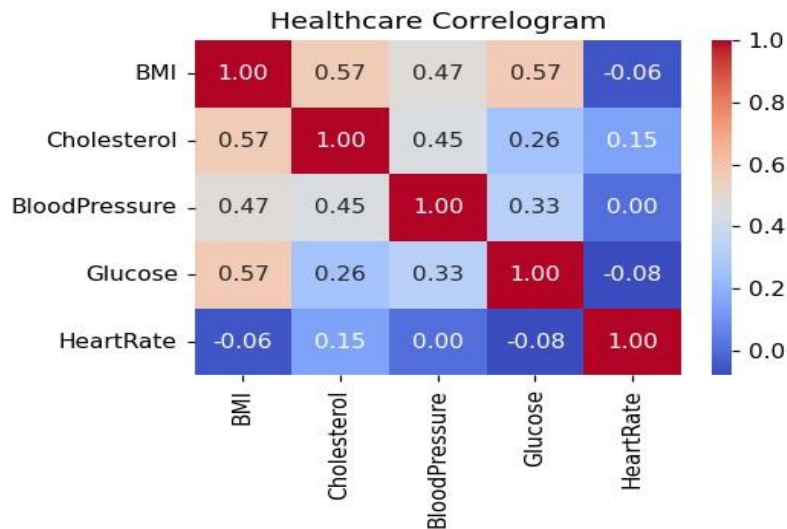
```
from sklearn.datasets import load_iris
import seaborn as sns, matplotlib.pyplot as plt
```

```
iris = load_iris(as_frame=True) X
= iris.data
sns.heatmap(X.corr(), annot=True, cmap='viridis', fmt=".2f") plt.title("Iris Correlation Heatmap") plt.show()
```



Q7. Healthcare Dataset — Correlogram & Multicollinearity

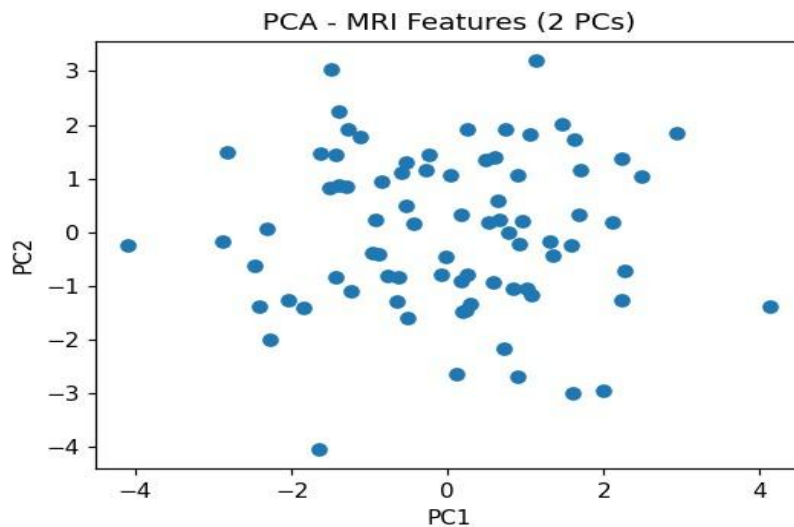
```
sns.heatmap(health_df.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Healthcare Correlogram") plt.show()
```



Q8. PCA on 25 MRI Features → 2 Components

```
from sklearn.decomposition import PCA
pca = PCA(n_components=2)
scores = pca.fit_transform(mri_features)  # (n_samples, 2)

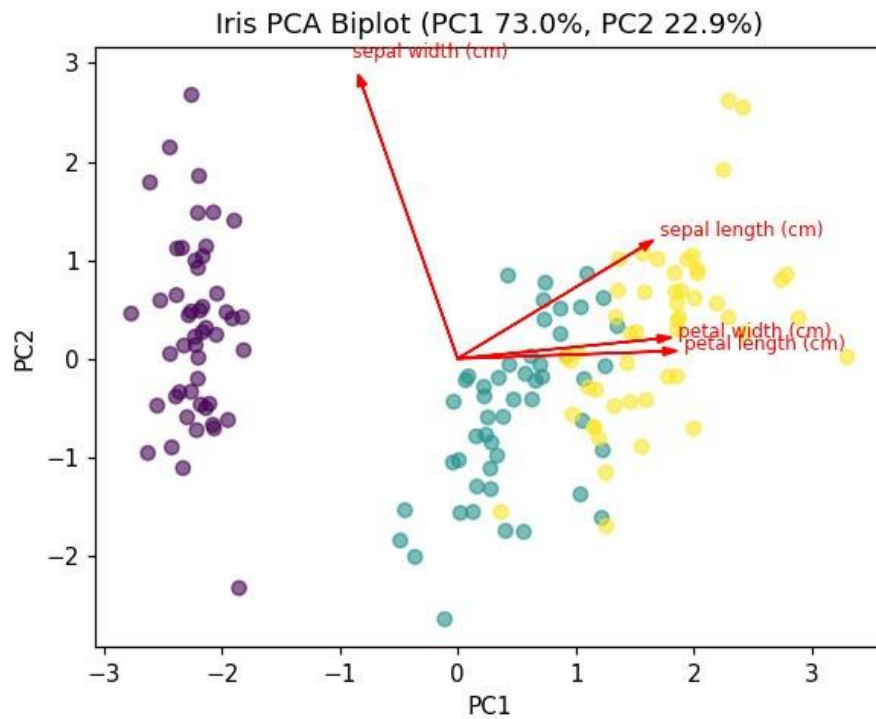
plt.scatter(scores[:,0], scores[:,1])
plt.xlabel("PC1"); plt.ylabel("PC2")
plt.title("PCA - MRI Features") plt.show()
```



Q9. Iris — PCA, Summary & Biplot

```
from sklearn.decomposition import PCA

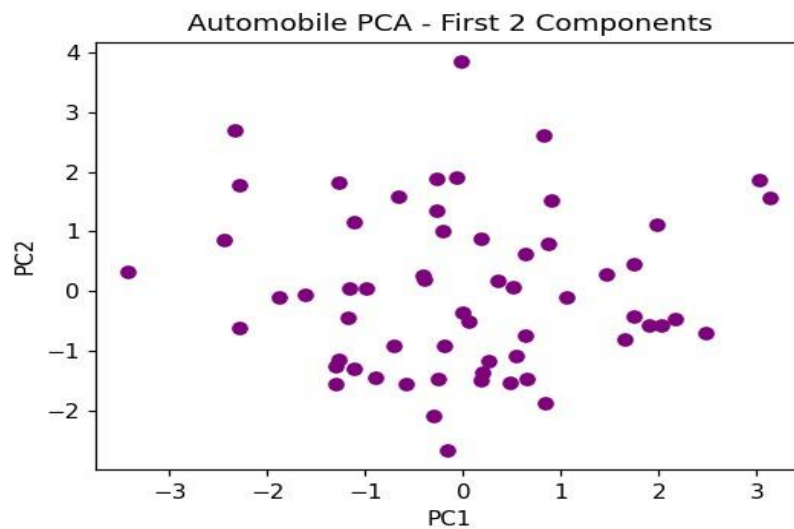
pca = PCA(n_components=2)
scores = pca.fit_transform(X_scaled)
print(pca.explained_variance_ratio_)
# biplot: scatter of scores + arrows for loadings
```



Q10. Automobile Specs (15 features) — PCA

```
pca = PCA(n_components=2)
scores = pca.fit_transform(auto_specs) # (n_cars, 15)

plt.scatter(scores[:,0], scores[:,1], color='purple')
plt.title("Automobile PCA - First 2 Components") plt.show()
```

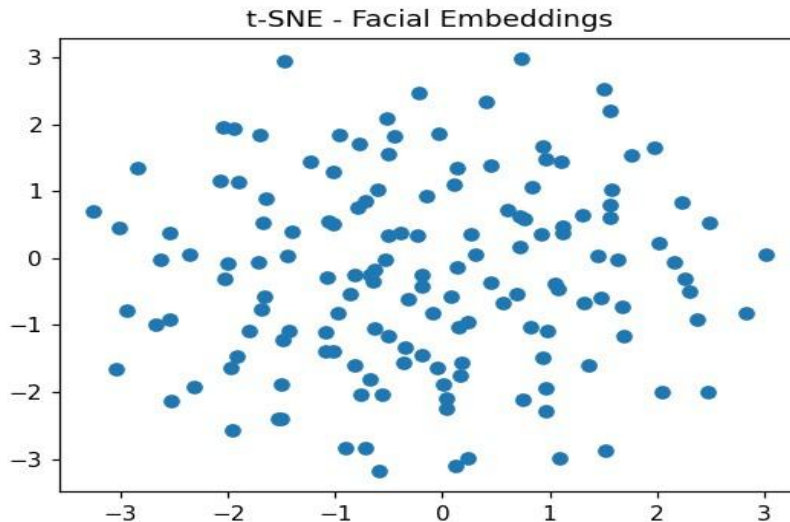


Q11. Facial Embeddings (128-D) — t-SNE

Task: Visualize with t-SNE; explain why t-SNE is preferred over PCA here.

```
from sklearn.manifold import TSNE
```

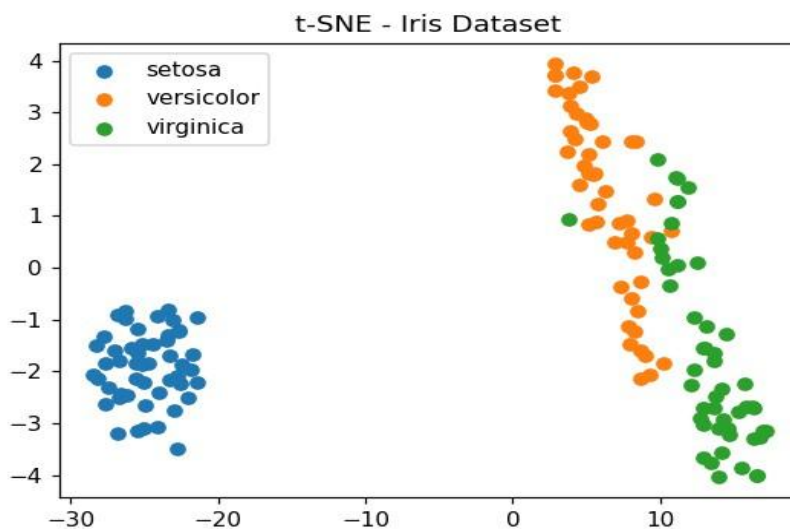
```
embed_2d = TSNE(n_components=2, perplexity=30, random_state=42).fit_transform(face_embeddings)
plt.scatter(embed_2d[:,0], embed_2d[:,1]) plt.title("t-SNE - Facial Embeddings") plt.show()
```



Interpretation: t-SNE is preferred over PCA for high-dimensional embeddings because it preserves local neighborhood structure (which faces are similar) rather than just directions of maximum variance, revealing nonlinear clusters that PCA's linear projection would blur together.

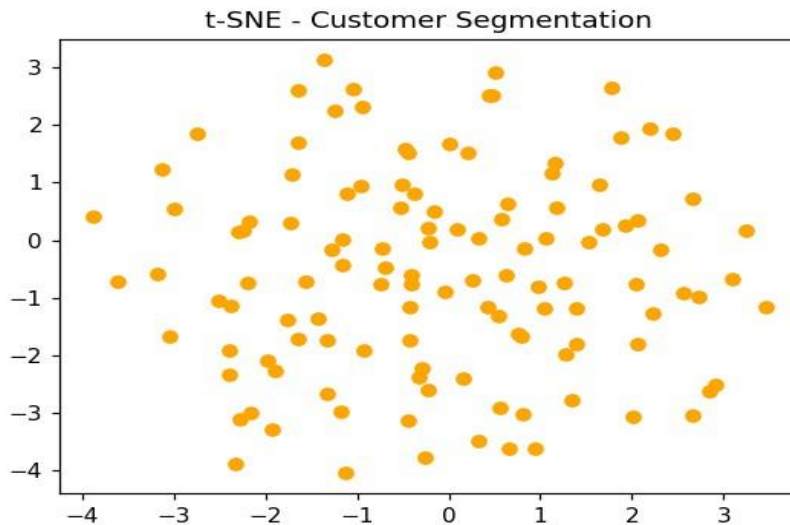
Q12. Iris — t-SNE Clusters by Species

```
tsne = TSNE(n_components=2, perplexity=30, random_state=42) embed = tsne.fit_transform(X)
for i, name in enumerate(iris.target_names):
    idx = iris.target == i
    plt.scatter(embed[idx,0], embed[idx,1], label=name)
plt.legend(); plt.title("t-SNE - Iris Dataset") plt.show()
```



Q13. Customer Segmentation (40 variables) — t-SNE

```
embed = TSNE(n_components=2, perplexity=30, random_state=42).fit_transform(customer_data)
plt.scatter(embed[:,0], embed[:,1], color='orange') plt.title("t-SNE - Customer Segmentation") plt.show()
```

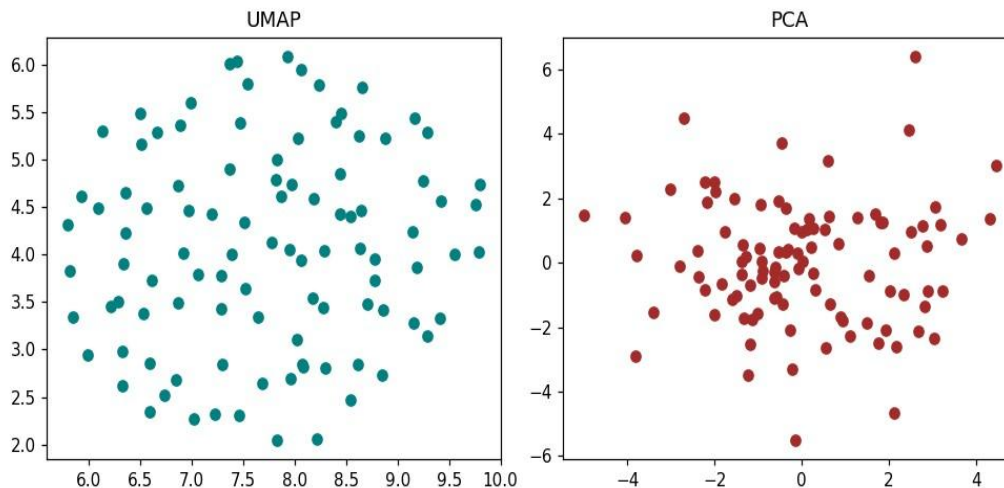


Q14. Medical Imaging (100 features) — UMAP vs PCA

```
import umap
from sklearn.decomposition import PCA

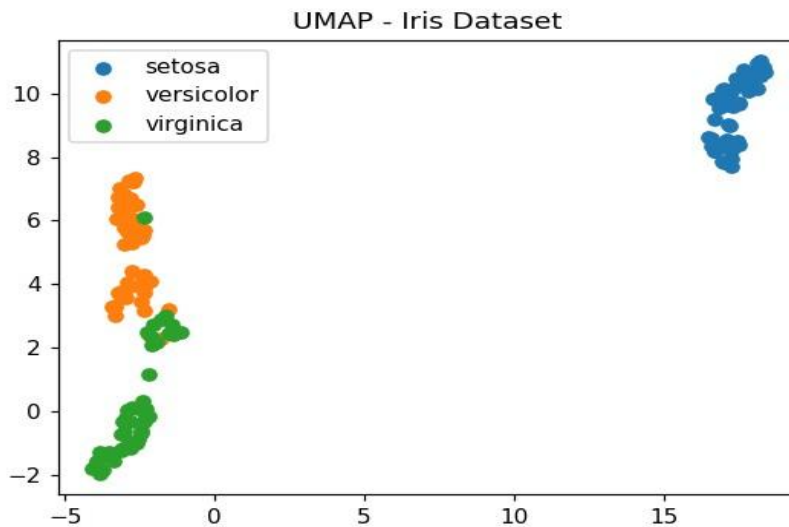
umap_embed = umap.UMAP(random_state=42).fit_transform(medical_features)
pca_embed = PCA(n_components=2).fit_transform(medical_features)

fig, axs = plt.subplots(1,2, figsize=(9,4))
axs[0].scatter(umap_embed[:,0], umap_embed[:,1]); axs[0].set_title("UMAP")
axs[1].scatter(pca_embed[:,0], pca_embed[:,1]); axs[1].set_title("PCA")
plt.show()
```



Q15. Iris — UMAP by Species

```
import umap
embed = umap.UMAP(random_state=42).fit_transform(X)
for i, name in enumerate(iris.target_names):
    idx = iris.target == i
    plt.scatter(embed[idx,0], embed[idx,1], label=name)
plt.legend(); plt.title("UMAP - Iris Dataset")
plt.show()
```



Q16. E-commerce Purchasing Behavior (60 attributes) — UMAP

```
embed = umap.UMAP(random_state=42).fit_transform(ecommerce_data)
plt.scatter(embed[:,0], embed[:,1], color='crimson')
plt.title("UMAP - E-commerce Customer Clusters") plt.show()
```

